

SQA & Testing

Chapter 3

Test planning and Design

The testing process

- Planning, design and performance of testing are carried out throughout the software development process.
- These activities are divided in phases, beginning in the design stage and ending when the software is installed at the customer's site.

Determining the test methodology



Planning the tests



Designing the tests



Performing the tests
(implementation)

Preparing test cases

- Estimating the number of new test cases needed for the current release
- Allocating the number of new test cases among the systems to be tested
- Allocating the number of new test cases for each system among its new operations
- Specify the new test cases
- Adding the new test cases to the test cases from previous releases.

ESTIMATING THE NUMBER OF NEW TEST CASES NEEDED

- Two factors must be taken into account:
 - Time
 - Cost
- Compute the number of test cases you have time to prepare
- Compute the number of test cases you can afford to prepare
- Take the minimum of these as the number of test cases you will plan to prepare

Test planning documentation

- The planning stage of the software system tests is commonly documented in a “software test plan” (STP)

STP

1 Scope of the tests

- 1.1 The software package to be tested (name, version and revision)
- 1.2 The documents that provide the basis for the planned tests (name and version for each document)

2 Testing environment

- 2.1 Testing sites
- 2.2 Required hardware and firmware configuration
- 2.3 Participating organizations
- 2.4 Manpower requirements
- 2.5 Preparation and training required of the test team

3 Test details (for each test)

- 3.1 Test identification
- 3.2 Test objective
- 3.3 Cross-reference to the relevant design document and the requirement document
- 3.4 Test class
- 3.5 Test level (unit, integration or system tests)
- 3.6 Test case requirements
- 3.7 Special requirements (e.g., measurements of response times, security requirements)
- 3.8 Data to be recorded

4 Test schedule (for each test or test group) including time estimates for the following:

- 4.1 Preparation
- 4.2 Testing
- 4.3 Error correction
- 4.4 Regression tests

...Test planning documentation

- The test design is carried out on the basis of the software test plan as documented by STP.
- The test procedures and the test case database/file may be documented in a “software test procedure” document and “test case file” document or in a single document called the “software test description” (STD)

STD

1 Scope of the tests

- 1.1 The software package to be tested (name, version and revision)
- 1.2 The documents providing the basis for the designed tests (name and version for each document)

2 Test environment (for each test)

- 2.1 Test identification (the test details are documented in the STP)
- 2.2 Detailed description of the operating system and hardware configuration and the required switch settings for the tests
- 2.3 Instructions for software loading

3 Testing process

- 3.1 Instructions for input, detailing every step of the input process
- 3.2 Data to be recorded during the tests

4 Test cases (for each case)

- 4.1 Test case identification details
- 4.2 Input data and system settings
- 4.3 Expected intermediate results (if applicable)
- 4.4 Expected results (numerical, message, activation of equipment, etc.)

5 Actions to be taken in case of program failure/cessation

6 Procedures to be applied according to the test results summary

Test case specification

- The IEEE standard 829-1983 provides templates for test case specifications and test procedure specifications.
- A test case specification consists of the following components:
 - test case specification identifier,
 - test items,
 - input specification,
 - output specification,
 - special environment needs,
 - special procedural requirements, and
 - test case dependencies.
- A test procedure specification consists of the following components:
 - test procedure specification identifier,
 - purpose,
 - specific requirements, and
 - procedure steps

TEST PLANNING AND DESIGN

TEST PLANNING

- The purpose of system test planning, or simply test planning, is to get ready and organized for test execution.
- A test plan provides a framework, scope, details of resource needed, effort required, schedule of activities, and a budget.
- A framework is a set of ideas, facts, or circumstances within which the tests will be conducted. The stated scope outlines the domain, or extent, of the test activities.
- The scope covers the managerial aspects of testing, rather than the detailed techniques and specific test cases.

...TEST PLANNING AND DESIGN

TEST DESIGN

- is a critical phase of software testing. During the test design phase, the system requirements are critically studied, system features to be tested are thoroughly identified, and the objectives of test cases and the detailed behavior of test cases are defined.
- Test objectives are identified from different sources, namely, the requirement specification and the functional specification, and one or more test cases are designed for each test objective.
- Each test case is designed as a combination of modular test components called test steps. These test steps can be combined together to create more complex, multistep tests
- A test case is clearly specified so that others can easily borrow, understand, and reuse it.

...TEST PLANNING AND DESIGN

- It is interesting to note that a new test-centric approach to system development is gradually emerging. This approach is called test-driven development (TDD)
- In test-driven development, programmers design and implement test cases before the production code is written.
- This approach is a key practice in modern agile software development processes.
- The main characteristics of agile software development processes are:
 - (i) incremental development,
 - (ii) coding of unit and acceptance tests conducted by the programmers along with customers,
 - (iii) frequent regression testing, and
 - (iv) writing test code, one test case at a time, before the production code.

Test team organization

- Test groups
- It is recommended to have at least two test groups:
 - integration test group and
 - system test group.

...Test team organization

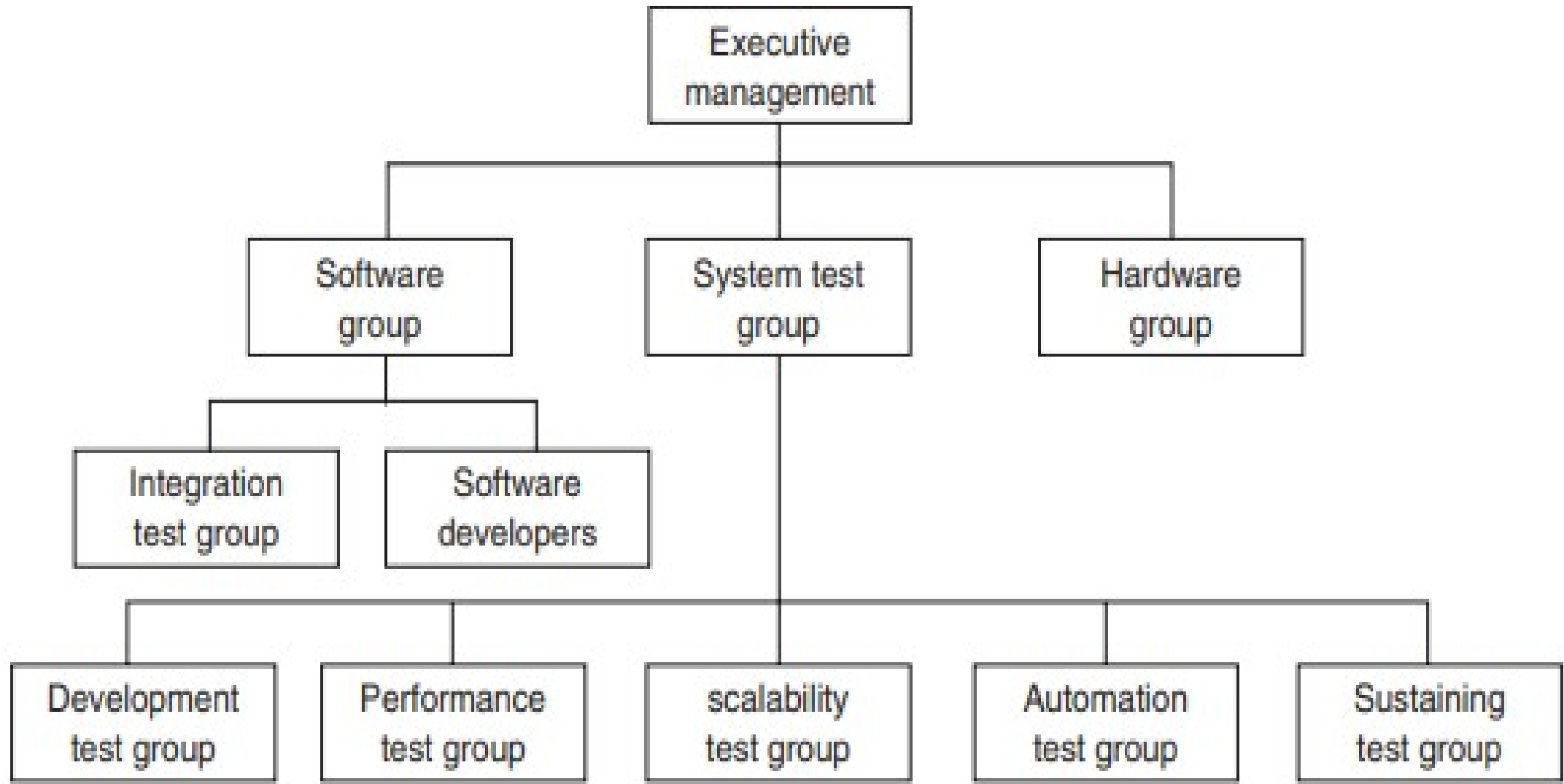
Integration Test Group

- Integration testing, is performed by an integration test group.
- Integration testing requires a thorough knowledge of the modules and their interfaces.
- Software developers, who together built the modules, must be involved in performing integration testing.
- In practice, the developers themselves may integrate the system. The system architects are also involved in integration testing for complex systems because of their thorough understanding of the larger picture of the system
- In addition to integrating the system, the test group may perform other duties, such as code inspection, configuration management, release management, and management of development laboratory.

...Test team organization

System Test Group

- A system test group is a team whose primary job is to perform testing without any bias and fear.
- The system test group is truly an independent group, and it usually has a separate head count (people that are specifically hired and retained to perform testing) and budget.
- The mandate of this group is to ensure that the system requirements have been satisfied and that the system is acceptable.
- The system test group can be further divided into several focused subgroups as the size of an organization becomes large: development, performance, scalability, automation, and sustaining test group.



...System Test Group

- Development Test Group

testing of new features in a particular release; basic tests, functionality tests, robustness tests, interoperability tests, stress tests, load and stability tests, regression tests, documentation tests, and business acceptance tests.

- Performance Test Group

emphasis on system performance. Tests are conducted to identify system bottlenecks and recommendations are made to the developers for improving system performance.

...System Test Group

- Scalability Test Group

determining whether or not the system can scale up to its engineering limits.

For example, a cellular phone network might have been designed with certain engineering limits in mind, such as the maximum number of base stations it can support, the maximum number of simultaneous calls it can handle, and so on. The group tests whether the designed system can reach those limits. This group may take up additional responsibilities such as load and stability testing.

...System Test Group

- Automation Test Group

The responsibility of this group is to develop test automation infrastructure, test libraries, and tests tools.

This group assists other groups in the development of automated test suites.

- Sustaining Test Group

This group maintains the software quality throughout the product's market life.

This team is responsible for maintaining the corrective aspect of software maintenance.

The group works very closely with customers and conducts regression testing of patch software.